

CD — Colour by Design

OCR B (Salters) H433 storyline **CD**. Dyes and colour to order: the origins of colour in delocalised systems, arenes and their reactions, diazonium compounds and dyeing, fats and oils, gas-liquid chromatography and carbonyl chemistry.

1. Aromatic Chemistry

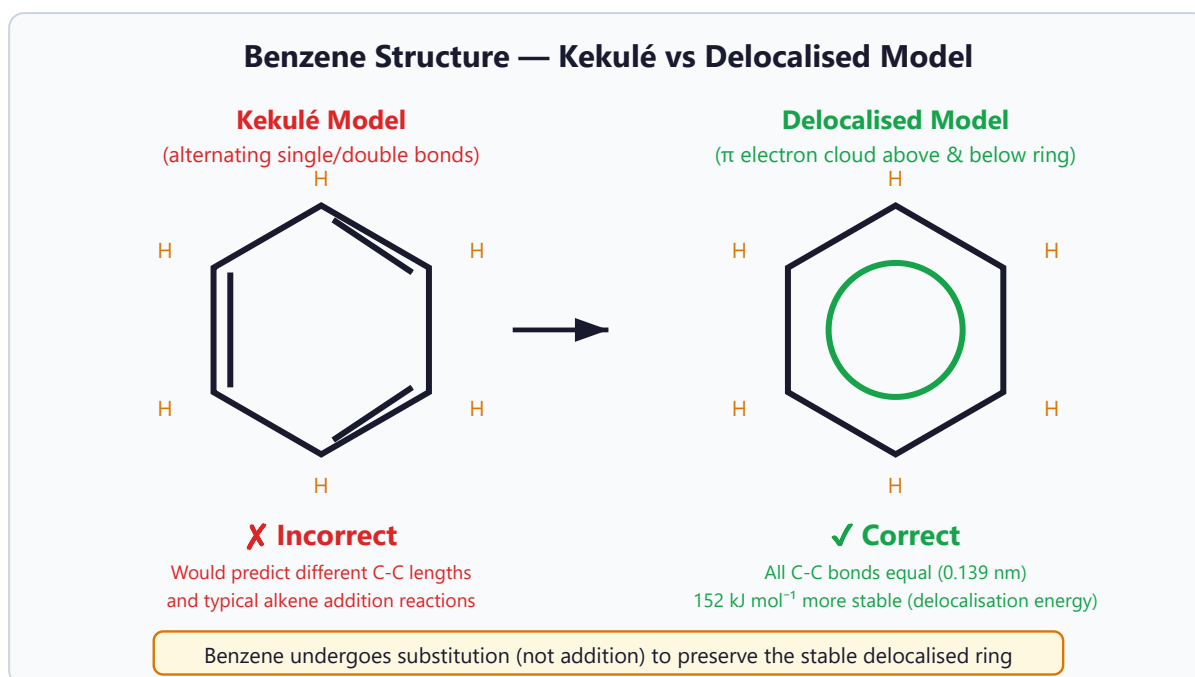
Structure of Benzene

Benzene (C_6H_6): A planar, cyclic hydrocarbon with a ring of six carbon atoms. Each carbon is sp^2 hybridised with bond angles of 120° . The unhybridised p orbital on each carbon overlaps sideways with those on adjacent carbons, forming a delocalised π system above and below the plane of the ring.

Kekulé model: Originally proposed as alternating single and double bonds (cyclohexatriene). However, this model is inconsistent with experimental evidence:

- All C–C bond lengths in benzene are equal (0.139 nm), intermediate between a single bond (0.154 nm) and a double bond (0.134 nm).
- Benzene does not undergo addition reactions typical of alkenes (e.g. it does not readily decolourise bromine water).
- The enthalpy of hydrogenation of benzene (-208 kJ mol^{-1}) is less exothermic than predicted for three C=C bonds ($3 \times -120 = -360 \text{ kJ mol}^{-1}$), showing benzene is more stable than the hypothetical cyclohexatriene by about 152 kJ mol^{-1} (the delocalisation/resonance energy).

Delocalised model: The six π electrons are spread evenly over all six carbon atoms in a ring of electron density. This delocalisation makes benzene more thermodynamically stable than the Kekulé structure and explains its tendency to undergo substitution rather than addition reactions (preserving the stable delocalised ring).



Electrophilic Substitution of Benzene

The delocalised π electron ring is a region of high electron density that attracts electrophiles. However, benzene undergoes substitution rather than addition to preserve the stability of the aromatic ring.

Nitration:



The electrophile is the nitronium ion (NO_2^+), generated by the reaction of concentrated nitric acid with concentrated sulfuric acid: $\text{HNO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + \text{HSO}_4^- + \text{H}_2\text{O}$. The mechanism involves the NO_2^+ being attracted to the π system, forming a positively charged intermediate (Wheland intermediate), followed by loss of H^+ to restore aromaticity. Temperature must be kept below 55°C to prevent dinitration.

Halogenation:



A halogen carrier catalyst (Lewis acid such as AlBr_3 , FeBr_3 , or AlCl_3) is needed to generate the electrophile. The catalyst polarises the $\text{Br}-\text{Br}$ bond or forms Br^+ : $\text{Br}_2 + \text{AlBr}_3 \rightarrow \text{Br}^+ + \text{AlBr}_4^-$. The Br^+ then attacks the π ring.

Friedel-Crafts alkylation:



The AlCl_3 catalyst generates a carbocation electrophile (R^+) from the halogenoalkane. This introduces an alkyl group onto the benzene ring.

Friedel-Crafts acylation:



The electrophile is an acylium ion (RCO^+). This introduces an acyl group ($\text{C}=\text{O}$ bonded to R) onto the ring. Acylation is preferred over alkylation in synthesis because it does not give multiple substitutions (the $\text{C}=\text{O}$ group deactivates the ring).

Exam Tip: In all electrophilic substitution mechanisms on benzene, draw two curly arrows: one from the π ring to the electrophile (forming the new bond), and one from the $\text{C}-\text{H}$ bond to the ring (restoring aromaticity by losing H^+). Show the intermediate Wheland complex with its positive charge delocalised over the ring.

2. Carbonyl Compounds

Aldehydes and Ketones

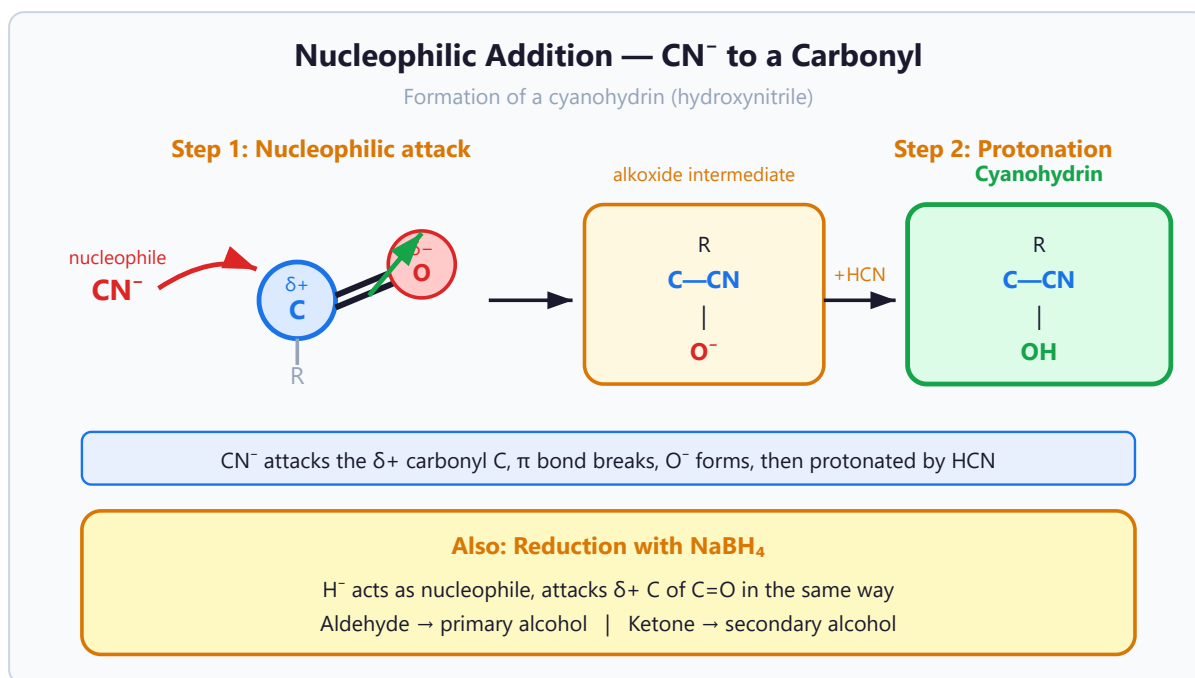
Both contain the $\text{C}=\text{O}$ (carbonyl) group. Aldehydes have the carbonyl at the end of the chain (RCHO); ketones have it within the chain (RCOR'). The $\text{C}=\text{O}$ bond is polar (δ^+ on C , δ^- on O) because oxygen is more electronegative than carbon, making the carbonyl carbon susceptible to nucleophilic attack.

Nucleophilic Addition

The key mechanism for carbonyl compounds. A nucleophile attacks the δ^+ carbon of the $\text{C}=\text{O}$ group:

With HCN (hydrogen cyanide): The nucleophile is CN^- . It attacks the δ^+ carbon, breaking the π bond and forming an alkoxide intermediate, which is then protonated by HCN to give a hydroxynitrile (cyanohydrin). This reaction extends the carbon chain by one carbon atom and introduces both an OH and CN group. A trace of base (NaCN or KCN) is used as a catalyst to generate CN^- .

With NaBH₄ (sodium borohydride): A reducing agent. The H⁻ ion acts as the nucleophile, attacking the δ⁺ carbonyl carbon. Aldehydes are reduced to primary alcohols; ketones are reduced to secondary alcohols. NaBH₄ is used in aqueous/ethanolic solution.



Distinguishing Aldehydes from Ketones

Test	Reagent	Aldehyde result	Ketone result
Tollens' test	[Ag(NH ₃) ₂] ⁺ (silver mirror reagent), warm	Silver mirror forms (Ag ⁺ reduced to Ag)	No reaction
Fehling's test	Fehling's solution (Cu ²⁺ tartrate complex), heat	Brick-red precipitate of Cu ₂ O	No reaction
2,4-DNP test	2,4-dinitrophenylhydrazine (Brady's reagent)	Orange/yellow precipitate (confirms C=O present)	Orange/yellow precipitate (confirms C=O present)

Tollens' and Fehling's tests work because aldehydes can be oxidised to carboxylic acids, but ketones cannot be oxidised under these mild conditions. The 2,4-DNP test confirms the presence of a carbonyl group (aldehyde or ketone) but does not distinguish between them. The melting point of the 2,4-DNP derivative can be used to identify the specific aldehyde or ketone.

